

Logarithmic Scale Weighted Cross-Entropy for Class-Imbalanced Learning

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- ▶ Class imbalance: Common in healthcare, finance, manufacturing.
- ▶ Models on imbalanced data: High accuracy by ignoring minorities → deceptive.
- ▶ Fatal risks: Missed rare diseases, undetected fraud, overlooked defects.

`confusion_matrix_v3.pdf`

1. Data Resampling

- ▶ **What it is:** Modifies the training set distribution to achieve a better balance.
- ▶ **Examples:**
 - ▶ Oversampling (SMOTE, ADASYN)
 - ▶ Undersampling (Tomek Links, ENN)

2. Cost-Sensitive Learning

- ▶ **What it is:** Adjusts the algorithm to assign a higher penalty for misclassifying minority samples.
- ▶ **Examples:**
 - ▶ Weighted Cross-Entropy (WCE)
 - ▶ Focal Loss, Class-Balanced Loss

Standard Cross-Entropy:

$$\mathcal{L}_{CE} = - \sum_{i=1}^N y_i \log(p_i)$$

Problem: The total loss is **dominated** by the majority class, and the gradient signal from the minority class is overwhelmed.

→ *Focus on* **Cost-Sensitive Learning**.

Existing Methods (I): Inverse Weighting Schemes

The simplest cost-sensitive method is to assign a weight w_i to the standard CE loss:

$$\mathcal{L}_{WCE} = - \sum_{i=1}^N w_i \cdot y_i \log(p_i)$$

Weighted Cross-Entropy (WCE) & Inverse Sqrt WCE

- ▶ **Core Idea:** Weights are inversely proportional to class frequency, sometimes softened.
- ▶ **Weight Formula** (n_i : samples in class i):

$$w_i \propto \frac{1}{n_i} \quad \text{or} \quad w_i \propto \frac{1}{\sqrt{n_i}}$$

1. Focal Loss

- ▶ **Core Idea:** Focuses on **sample difficulty** instead of class frequency.
- ▶ Down-weights the loss of "easy" (often majority) samples.
- ▶ **Formula** (p_t : prob. of true class):

$$\mathcal{L}_{FL} = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

- ▶ **Key Hyperparameter:**
 - ▶ γ controls the focusing rate.

2. Class-Balanced (CB) Loss

- ▶ **Core Idea:** A principled re-weighting based on the "effective number of samples".
- ▶ **Weight Formula** ($N_i(n_i, \beta)$):

$$w_i = \frac{1 - \beta}{1 - \beta^{n_i}}$$

$$\mathcal{L}_{CB} = - \sum w_i \cdot \mathcal{L}_{CE}(y_i, p_i)$$

- ▶ **Key Hyperparameter:**
 - ▶ β controls the curve.

Proposed Methodology (I): The Core Idea

Motivation: Instability of WCE

Standard WCE weight:

$$w_i = \frac{N_{total}}{n_i} \propto \frac{1}{n_i}$$

Critical Issue

Severe imbalance ($n_i \rightarrow$ small)
causes w_i to **explode**.

- ▶ Numerical instability.
- ▶ Severe overfitting to minority

Solution: Log-Weighted CE (LWCE)

We use log-scaling to **tame** the weights:

$$w_i \propto \frac{1}{\log(n_i + 1)}$$

wce_vs_lwce_relative_v2.pdf

Generalization: Power-Scaled Adaptive CE (PLWCE)

We introduce α to flexibly control the **focusing strength**:

$$w_i \propto \frac{1}{(\log(n_i + 1))^\alpha}$$

Alpha (α)	Role & Effect
$\alpha = 1.0$	Balanced LWCE
$\alpha > 1.0$	Aggressive focus on minority
$\alpha < 1.0$	Conservative weighting

Table: Role of the α parameter

plwce_alpha_relative_zoomed.pdf

Table: Benchmark Datasets (Sorted by IR)

Dataset	# Classes	Imbalance Ratio (IR)
Credit Card Fraud	2	587:1
APS Failure	2	59:1
Page Blocks	5	175:1
Yeast	10	93:1
... (Total 11)	...	...
German Credit	2	2.3:1

Base Model: XGBoost

XGBoost with linear booster ('gblinear') as a robust baseline.

- ▶ Handles both binary and multi-class tasks.
- ▶ Efficient for large-scale tabular data.

Evaluation Metrics

Standard accuracy is misleading for imbalanced data. We focus on:

- ▶ **F1-Score:** Harmonic mean of precision and recall, averaged across classes.
- ▶ **PR-AUC:** Area under the Precision-Recall curve, highly sensitive to minority class performance.

Hyperparameter Optimization

We used **Optuna** for automated hyperparameter tuning.

- ▶ **Search Space:**
 - ▶ learning_rate, n_estimators
 - ▶ Regularization: lambda (L2), alpha (L1)
 - ▶ Loss-specific: γ (Focal), β (CB), α (PS-WCE, PLWCE)
- ▶ **Protocol:** 3-fold CV, 100 trials per dataset.

Full Experimental Results (I)

Credit Card Fraud

Loss	F1	PR
CE	0.722	0.711
WCE	0.796	0.698
WCE(sq)	0.797	0.700
PS-WCE	0.799	0.701
Focal	0.757	0.715
CB	0.801	0.704
LWCE	0.747	0.716
PLWCE	0.806	0.701

Aps Failure

Loss	F1	PR
CE	0.759	0.812
WCE	0.648	0.813
WCE(sq)	0.761	0.825
PS-WCE	0.776	0.825
Focal	0.774	0.826
CB	0.756	0.827
LWCE	0.765	0.824
PLWCE	0.773	0.828

Secom

Loss	F1	PR
CE	0.167	0.141
WCE	0.276	0.167
WCE(sq)	0.172	0.137
PS-WCE	0.189	0.136
Focal	0.145	0.139
CB	0.128	0.142
LWCE	0.222	0.177
PLWCE	0.213	0.171

Bank Marketing

Loss	F1	PR
CE	0.456	0.548
WCE	0.562	0.539
WCE(sq)	0.574	0.545
PS-WCE	0.573	0.545
Focal	0.567	0.546
CB	0.474	0.503
LWCE	0.489	0.548
PLWCE	0.558	0.546

Credit Card Default

Loss	F1	PR
CE	0.357	0.498
WCE	0.501	0.449
WCE(sq)	0.496	0.483
PS-WCE	0.483	0.479
Focal	0.510	0.48
CB	0.487	0.472
LWCE	0.488	0.473
PLWCE	0.484	0.479

Telco Churn

Loss	F1	PR
CE	0.593	0.628
WCE	0.623	0.643
WCE(sq)	0.629	0.629
PS-WCE	0.624	0.641
Focal	0.623	0.635
CB	0.630	0.626
LWCE	0.629	0.625
PLWCE	0.629	0.624

Full Experimental Results (II)

German Credit

Loss	F1	PR
CE	0.585	0.640
WCE	0.630	0.631
WCE(sq)	0.615	0.644
PS-WCE	0.602	0.639
Focal	0.601	0.641
CB	0.626	0.641
LWCE	0.591	0.641
PLWCE	0.628	0.642

Page Blocks (Multi)

Loss	F1	PR
CE	0.829	0.885
WCE	0.006	0.200
WCE(sq)	0.835	0.883
PS-WCE	0.813	0.882
Focal	0.792	0.843
CB	0.838	0.883
LWCE	0.815	0.892
PLWCE	0.834	0.893

Yeast (Multi)

Loss	F1	PR
CE	0.577	0.597
WCE	0.447	0.570
WCE(sq)	0.521	0.593
PS-WCE	0.533	0.593
Focal	0.462	0.514
CB	0.518	0.592
LWCE	0.541	0.598
PLWCE	0.528	0.594

Steel Faults (Multi)

Loss	F1	PR
CE	0.706	0.726
WCE	0.628	0.721
WCE(sq)	0.691	0.725
PS-WCE	0.698	0.726
Focal	0.667	0.707
CB	0.706	0.725
LWCE	0.710	0.726
PLWCE	0.696	0.726

Glass (Multi)

Loss	F1	PR
CE	0.661	0.731
WCE	0.668	0.695
WCE(sq)	0.707	0.735
PS-WCE	0.719	0.705
Focal	0.698	0.744
CB	0.710	0.736
LWCE	0.736	0.740
PLWCE	0.678	0.719

Overall Performance Summary: Heatmap

f1_heatmap.pdf

pr_auc_heatmap.pdf

Key Finding: Robustness

LWCE & **PLWCE** are consistently top-ranked (Y-axis). The **green colors** in the **top-left region** confirm their robustness on datasets with extreme imbalance.

Observation from Results

- ▶ The results show that LWCE ($\alpha = 1.0$) is already a very strong and robust baseline, often outperforming SOTA methods.
- ▶ However, in 5 out of 11 datasets, PLWCE (tuning $\alpha > 1.0$) achieved an even higher F1-Score than LWCE.

Conclusion: Alpha as a "Performance Booster"

α acts as a data-specific "booster":

- ▶ LWCE ($\alpha = 1.0$) is sufficient and provides top-tier performance.
- ▶ For some **extremely imbalanced** datasets, tuning $\alpha > 1$ allows PLWCE to find a better optimum, boosting **both F1 and PR-AUC** scores.

→ *PLWCE generalizes LWCE, providing an optional tuning parameter for achieving the best possible performance in critical scenarios.*

Challenge: Isolating α vs. IR

- ▶ **Ideal Experiment:** Fix all model parameters (LR, L1/L2, etc.) across all 11 datasets, and tune only α .
- ▶ **Problem:** A single set of fixed parameters failed. It was too sub-optimal for diverse datasets (e.g., `credit_card_fraud`), causing poor metrics and rendering the entire imbalanced data analysis meaningless under these conditions

`ir_vs_alpha_correlation.pdf`

- ▶ Shows a **weak positive trend**: higher imbalance (larger IR) seems to prefer a larger α ($\alpha > 1$).

1. Application to Other Models

- ▶ **Tree-based Models:** Validate LWCE/PLWCE on non-linear 'gbtree' models (standard XGBoost, LightGBM).
- ▶ **Deep Learning:** Apply to CNNs/Transformers for highly imbalanced tasks like object detection or medical segmentation.

2. Combining with Dynamic Sample Weighting

- ▶ Our method (PLWCE) uses **static, class-level** weights.
- ▶ **Hypothesis:** We can combine this with **dynamic, sample-level** methods (like Focal Loss or GHM) that focus on "easy" vs. "hard" samples.
- ▶ **Goal:** Investigate a hybrid loss that benefits from both stable class-based re-weighting (our α) and dynamic sample-level focus (like Focal's γ).

3. Automated Alpha (α) Tuning

- ▶ As discussed (P18), isolating α 's effect is challenging due to other interacting hyperparameters.
- ▶ **Controlled Experiment:**
 - ▶ Use a large dataset (e.g., `credit_card_fraud`).
 - ▶ Fix minority samples and **sub-sample the majority class** to create 5+ datasets with controlled IRs (e.g., 5:1, 10:1, 100:1).
- ▶ **Goal:** Find a clear correlation between **IR** and the **optimal α** by running alpha-only tuning on these controlled datasets.
- ▶ **Ultimate Aim:** Develop a heuristic or formula to set α **automatically**, eliminating the need for expensive trial-and-error tuning.

Key Contributions

1. **Identified Instability:** We highlighted the numerical instability of standard WCE ($1/n_i$) in extreme imbalance scenarios.
2. **Proposed LWCE:** We proposed **LWCE** ($1/\log(n_i + 1)$) as a simple, stable, and parameter-free solution that fundamentally tames weight explosion.
3. **Proposed PLWCE:** We generalized this into **PLWCE** ($1/(\log n_i)^\alpha$), demonstrating how α acts as a performance booster for extremely imbalanced datasets.
4. **Validated Performance:** Our methods proved **highly robust and competitive**, consistently ranking in the Top 2 across 11 diverse datasets against strong baselines.

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Thank You for Your Attention!

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